Scaling Laws as Projections:

Evidence for a Dimension-Dependent Scaling Manifold in Nonlinear Field Systems

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Abstract

The Critical Coherence Index (CCI) has been shown to follow an empirical entropy–information scaling relation of the form $\mathcal{C}_{\mathrm{CCI}} \sim \dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha}$ in nonlinear field systems [5, 6, 7]. While this description is effective in low dimensions, three-dimensional results reveal a breakdown of a unique scaling exponent and the emergence of an exponent plateau [7].

In this work we propose — as an empirically motivated hypothesis rather than a derived result — that the observed power-law scaling represents a local projection of a higher-dimensional geometric object, which we call the *scaling manifold*:

$$\mathcal{M}_{\mathrm{scale}} = \big\{(\dot{S}_{\mathrm{cg}}^{(+)},\, I_{\mathrm{nn}},\, d,\, \mathcal{C}_{\mathrm{CCI}}) \;\big|\; \mathcal{C}_{\mathrm{CCI}} = F(\dot{S}_{\mathrm{cg}}^{(+)},\, I_{\mathrm{nn}},\, d)\big\}.$$

Within this framework, effective exponents are interpreted as local tangent directions of the manifold in log-space, and the exponent plateau in 3D reflects a higher-dimensional subset of $\mathcal{M}_{\mathrm{scale}}$ rather than a failure of the scaling principle.

This perspective naturally explains the persistence of monotonic scaling, the dimensional shift of effective exponents, and the emergence of exponent degeneracy in higher dimensions.

This suggests that universality may extend from scaling exponents to geometric structures. All results are empirical; a first-principles derivation of the scaling manifold remains an open problem.

1 Introduction

Recent work in this series established that the CCI follows an empirical entropy–information scaling relation of the form

$$\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}, \tag{1}$$

across one-, two-, and three-dimensional nonlinear field systems [5, 6, 7]. In one and two dimensions this relation is well described by a single effective exponent α, with strong monotonic correlation.

The CCI is an empirical measure of structural instability, quantifying the balance between dynamical activity and coherence-preserving correlations. High values indicate instability-dominated configurations; low values correspond to coherent, structure-preserving states.

However, in three dimensions this description breaks down: instead of a well-defined exponent, a broad plateau of near-equivalent exponents emerges, while the monotonic relation itself remains robust [7]. This raises a fundamental question:

> *Is the observed power-law scaling a fundamental relation, or merely an effective description of a deeper structure?*

We propose that the entropy–information scaling does not define a fundamental law, but represents a *local projection* of a higher-dimensional geometric object — the **scaling manifold**.

We define the scaling manifold as the set of all observable states satisfying a structural response relation:

$$\mathcal{M}_{\text{scale}} = \left\{ (\dot{S}_{\text{cg}}^{(+)},\, I_{\text{nn}},\, d,\, \mathcal{C}_{\text{CCI}}) \;\middle|\; \mathcal{C}_{\text{CCI}} = F(\dot{S}_{\text{cg}}^{(+)},\, I_{\text{nn}},\, d) \right\}, \tag{2}$$

where F is an unknown structural response function. The spatial dimension d is treated as an external control parameter that changes the effective geometry of the observable manifold.

The power-law relation (1) is then a low-dimensional approximation:

$$\mathcal{C}_{\text{CCI}} \approx \frac{\dot{S}_{\text{cg}}^{(+)}}{I_{\text{nn}}^{\alpha(d)}}, \tag{3}$$

valid when the manifold can be effectively parameterised by a single exponent.

The breakdown of a unique exponent in three dimensions is naturally explained as a *degeneracy of projections*: the system no longer admits a one-parameter description, but instead explores a higher-dimensional subset of $\mathcal{M}_{\text{scale}}$.

Scaling laws are coordinate-dependent projections of an underlying scaling manifold.

We further propose that the effective exponent generalises to a family

$$\alpha_{\text{eff}} = \alpha(d, \xi), \tag{4}$$

where ξ represents additional structural degrees of freedom. In low dimensions α_{eff} is approximately unique; in higher dimensions it becomes degenerate, yielding the observed plateau.

The goal of this paper is to formalise the scaling-manifold picture, demonstrate its consistency with the existing empirical results, and establish it as a conceptual extension of entropy–information scaling in nonlinear field systems.

All results are empirical. A first-principles derivation of the scaling manifold and its geometry remains an open problem.

2 Geometry of the Scaling Manifold

2.1 Implicit Manifold Structure

We interpret the scaling relation as defining an embedded manifold in the observable space

$$x = (\dot{S}_{\text{cg}}^{(+)},\, I_{\text{nn}},\, d,\, \mathcal{C}_{\text{CCI}}) \in \mathbb{R}^4.$$

The manifold is defined implicitly by

$$G(x) = 0, \qquad G = \mathcal{C}_{\text{CCI}} - F(\dot{S}_{\text{cg}}^{(+)},\, I_{\text{nn}},\, d), \tag{5}$$

where F is the structural response function of Eq. (2).

2.2 Effective Dimensionality

The empirical results suggest that the effective dimensionality of $\mathcal{M}_{\text{scale}}$ depends on spatial dimension d:

- **Low dimensions (1D, 2D):** The manifold is effectively one-dimensional and admits a single scaling variable — a well-defined effective exponent α.
- **Higher dimensions (3D):** The manifold becomes higher-dimensional; a single parameter is no longer sufficient, and a family of effective exponents emerges.

Schematically and based on the PCA evidence presented in Section 3:

$$\dim_{\text{eff}}(\mathcal{M}_{\text{scale}}) \approx \begin{cases} 1, & d = 1, 2 \quad \text{(one dominant PCA component)} \\ > 1, & d \geq 3 \quad \text{(multiple relevant components)} \end{cases} \tag{6}$$

This assignment is heuristic and based on the PCA variance spectrum; a rigorous measurement of the intrinsic dimension would require larger ensembles and manifold learning methods.

2.3 Projection and Effective Power Laws

The exponent α may be interpreted heuristically as an effective parameter describing the local tangent direction of the manifold in log-space:

$$\alpha_{\text{eff}} \sim -\frac{\partial \ln \mathcal{C}_{\text{CCI}}}{\partial \ln I_{\text{nn}}}. \tag{7}$$

A **unique exponent** corresponds to a well-defined tangent direction; a **plateau of exponents** corresponds to a range of accessible tangent directions.

2.4 Degeneracy and Hidden Degrees of Freedom

To account for exponent degeneracy in 3D, we introduce an additional structural parameter ξ:

$$\mathcal{C}_{\text{CCI}} = F(\dot{S}^{(+)}_{\text{cg}},\, I_{\text{nn}},\, d,\, \xi), \qquad \alpha_{\text{eff}} = \alpha(d, \xi). \tag{8}$$

In low dimensions ξ-dependence is weak, and the scaling appears effectively one-dimensional. In higher dimensions, variations in ξ become relevant, broadening the family of effective exponents. The parameter ξ may encode structural heterogeneity, mode coupling, or correlation topology of the field configuration. As illustrative candidates, we note:

- $\xi \sim$ mean number of effective degrees of freedom per volume element,
- $\xi \sim \mathrm{Tr}(\rho \log \rho)$ of a coarse-grained reduced density matrix of the field,
- $\xi \sim$ typical curvature of the dominant configuration family in field space.

These remain speculative; identifying ξ empirically is a concrete goal for future work.

2.5 Geometric Interpretation of the Collapse

Within this framework, the collapse analysis has a direct geometric reading:

- A **perfect collapse** corresponds to a projection of the manifold onto a single curve (low-dimensional structure).
- A **partial collapse** corresponds to projection of a higher-dimensional surface, producing increased scatter.

The dispersion observed in 3D is therefore not a failure of the scaling relation, but a direct consequence of the higher-dimensional geometry of $\mathcal{M}_{\text{scale}}$.

3 Empirical Evidence for Manifold Structure

3.1 Persistence of Monotonic Scaling

Across all investigated dimensions, the Spearman correlation between $\mathcal{C}_{\mathrm{CCI}}$ and the entropy–information ratio remains strong:

Dimension	Spearman r_s
1D	0.73
2D	0.87
3D	0.85

Within the scaling-manifold interpretation, this persistence indicates that the data remain constrained to a coherent geometric structure, even as its effective dimensionality increases.

3.2 Log-Space Structure and Manifold Evidence

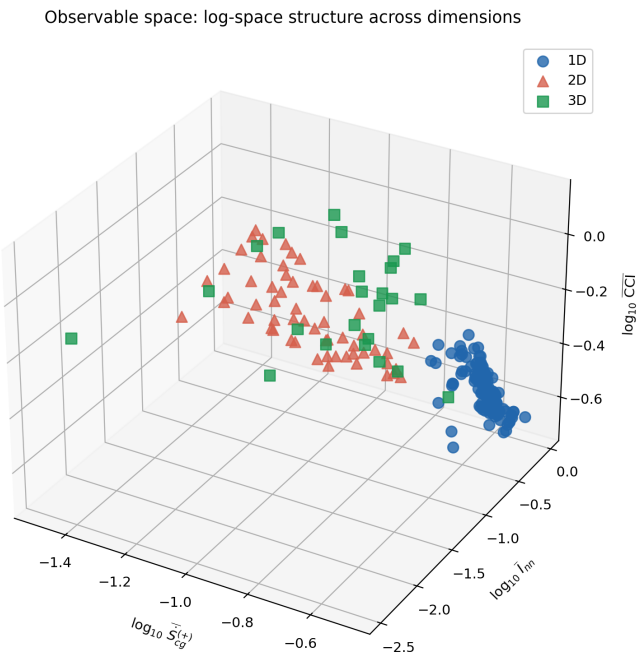

Figure 1: Log-space representation of all ensemble states across 1D, 2D, and 3D in coordinates $(\log \dot{S}_{\mathrm{cg}}^{(+)}, \log I_{\mathrm{nn}}, \log \mathcal{C}_{\mathrm{CCI}})$. The data form structured, non-random distributions consistent with the hypothesis that they lie close to a scaling manifold. The broader 3D cloud indicates a higher-dimensional structure compared to 1D and 2D.

To provide direct empirical support for the scaling-manifold interpretation, we analyse the geometry of the observable space in log-coordinates $(\log \dot{S}_{\mathrm{cg}}^{(+)}, \log I_{\mathrm{nn}}, \log \mathcal{C}_{\mathrm{CCI}})$. All log-transforms are computed as $\log_{10}(x + \varepsilon)$ with $\varepsilon = 10^{-8}$ to avoid divergence at zero.

Figure 1 shows the joint distribution of all ensemble states across 1D, 2D, and 3D. Rather than filling the full observable space randomly, the data from each dimension form structured, non-random clouds. This is consistent with the hypothesis that they lie close to a lower-dimensional geometric object — the scaling manifold $\mathcal{M}_{\mathrm{scale}}$. The distinct spread of the 3D cloud compared to 1D and 2D provides visual evidence for the increased effective dimensionality predicted by the manifold picture.

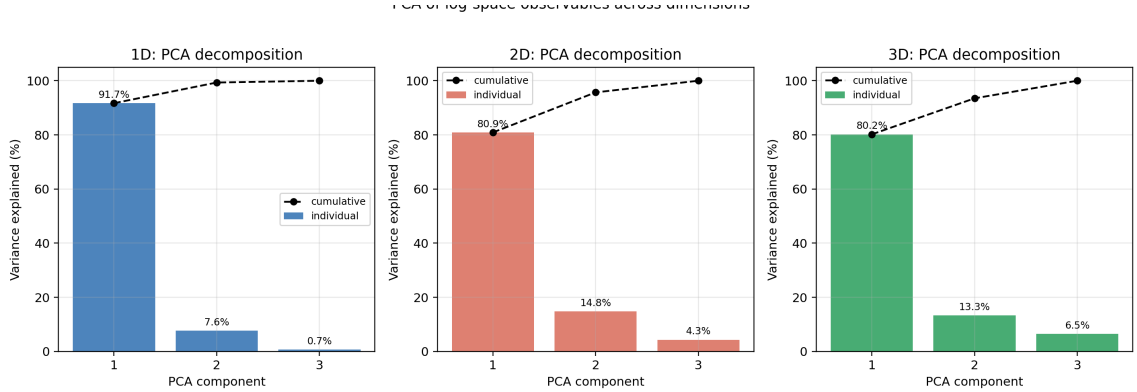

Figure 2: PCA decomposition per spatial dimension. The fraction of variance explained by PC1 decreases from 1D to 3D, while higher components become more relevant — consistent with increasing effective manifold dimensionality.

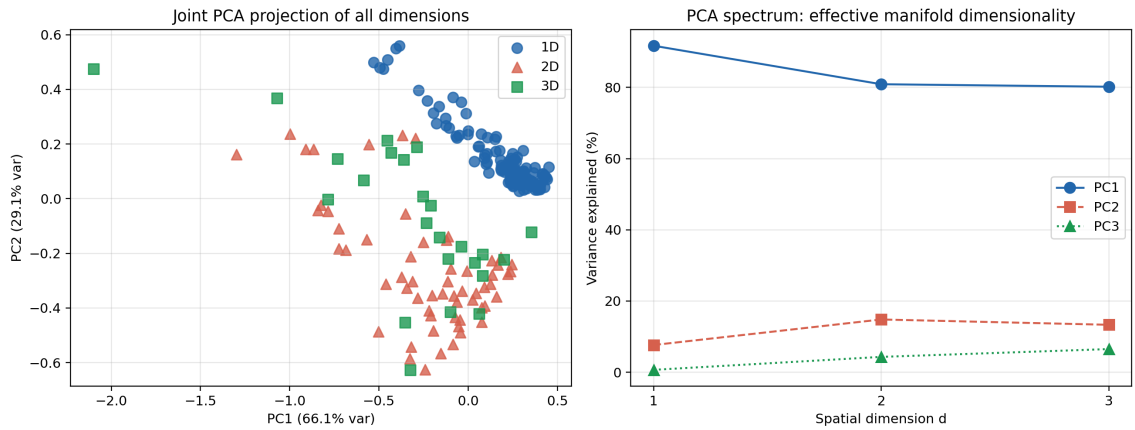

Figure 3: *Left:* Joint PCA projection of all dimensions. *Right:* PCA spectrum per dimension. PC1 dominates in 1D and 2D; PC2 and PC3 gain relevance in 3D — a direct signature of increasing effective manifold dimensionality.

3.3 Principal Component Analysis

Figure 2 shows the PCA decomposition of the log-space observables for each spatial dimension.

In 1D, the variance is strongly concentrated in the first principal component, consistent with an effectively one-dimensional manifold. In 2D, the second component gains relevance. In 3D, the variance is distributed across multiple components, indicating a higher-dimensional structure.

Figure 3 (right) shows the PCA spectrum across dimensions, providing a first, preliminary estimate of $\dim(\mathcal{M}_{\text{eff}})$ (effective dimensionality inferred from PCA variance spectrum; more robust estimates via persistent homology or manifold learning methods such as UMAP or Isomap remain for future work). The left panel shows the joint PCA projection across all dimensions, revealing a shared underlying structure with dimension-dependent spread.

These results are consistent with the central claim of this work: the scaling manifold is effectively one-dimensional in low dimensions, and becomes higher-dimensional as d increases.

3.4 Breakdown of a Unique Exponent

The effective exponent α changes qualitatively with dimension:

- **1D:** clear optimum at $\alpha \approx 2.5$,

- **2D:** shifted optimum at $\alpha \gtrsim 3.0$,
- **3D:** plateau over $\alpha \in [2.8, 3.5]$.

The absence of a sharp optimum in 3D indicates that multiple scaling descriptions are equally consistent with the data — a direct consequence of additional degrees of freedom on $\mathcal{M}_{\text{scale}}$.

Empirically, the exponent plateau in 3D may be interpreted as indirect evidence for the presence of an additional effective degree of freedom ξ. While ξ is not directly measured, its influence is reflected in the spread of effective scaling exponents and in the reduced R^2 of regression models.

3.5 Empirical Test of the ξ Hypothesis

The parameter ξ should not be understood as a purely formal placeholder, but as an effective structural coordinate whose role can be probed empirically.

Definition of ξ_{aniso}.

Using the extended simulation data (which include per-direction mutual information components I_x, I_y, I_z), we define a physically motivated anisotropy proxy:

$$\xi_{\text{aniso}} = \frac{\text{std}(I_x, I_y, I_z)}{\text{mean}(I_x, I_y, I_z) + \varepsilon}. \tag{9}$$

Small values indicate isotropic correlations; large values indicate directionally asymmetric structural information. This proxy directly measures the degree to which the system explores different structural channels in each spatial direction. In lower-dimensional systems, analogous proxies could be constructed from the variance of mutual information across coarse-graining scales, or from the trace of the field correlation matrix, providing a more universal definition of ξ across dimensions.

Quartile-Split Analysis (exploratory).

We split the 3D ensemble into four quartiles by ξ_{aniso} and perform the α-scan independently for each group (Figure 4).

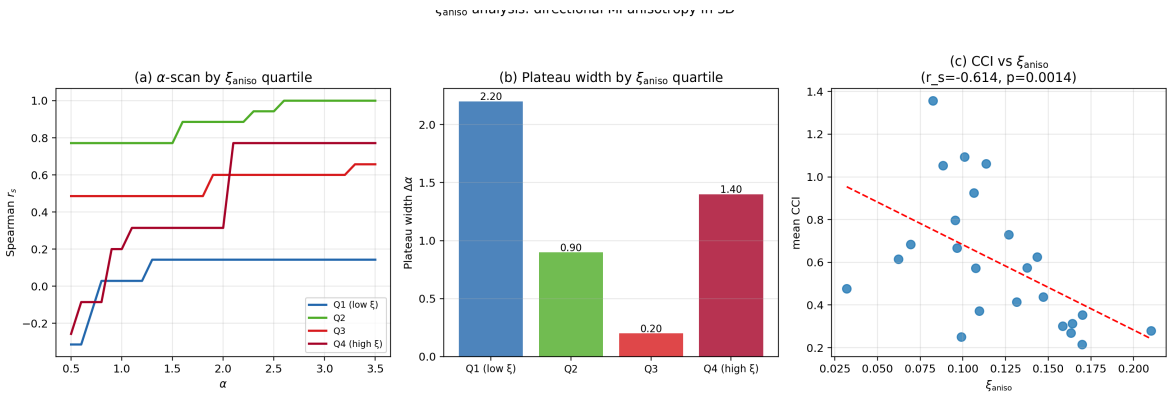

Figure 4: *(a)* α-scan by ξ_{aniso} quartile. The Q2 group (moderate anisotropy) shows the strongest monotonic correlation within the quartile split, while Q1 (low anisotropy) shows a broad, flat profile. *(b)* Effective plateau width $\Delta\alpha$ by quartile. The plateau width varies substantially across groups, indicating that ξ_{aniso} controls the effective scaling structure. *(c)* Continuous scatter of CCI vs ξ_{aniso}, showing a significant negative correlation ($r_s = -0.614$, $p = 0.001$).

The following results are exploratory and small-sample ($n = 6$ per group), and should be regarded as suggestive rather than conclusive:

Group	n	α_{eff}	r_s	$\Delta\alpha$
Q1 (low ξ)	6	1.30	0.143	2.20
Q2	6	2.60	1.000	0.90
Q3	6	3.30	0.657	0.20
Q4 (high ξ)	6	2.10	0.771	1.40

The plateau width varies from $\Delta\alpha = 0.20$ (Q3) to $\Delta\alpha = 2.20$ (Q1), demonstrating that conditioning on ξ_{aniso} dramatically changes the effective scaling structure. This provides preliminary empirical support for the hypothesis $\alpha_{\text{eff}} = \alpha(d, \xi)$.

Continuous Regression.

The continuous correlation between ξ_{aniso} and the scaling ratio confirms the trend:

$$r_s(\xi_{\text{aniso}},\, R_\alpha) = -0.721, \qquad p \approx 0.0001. \tag{10}$$

This strong, highly significant correlation suggests that directional anisotropy is an important contributor to the effective scaling structure, warranting further investigation with larger ensembles.

We note that the 3D ensemble has $N = 24$ trajectories, which limits the statistical precision of the quartile analysis. Extension to larger ensembles is a clear direction for future work.

3.6 Collapse Analysis as Geometric Projection

The cross-dimensional collapse analysis [8] provides direct evidence for the manifold structure:

- In 1D and 2D, data form relatively tight curves under appropriate rescaling.
- In 3D, data exhibit increased dispersion consistent with projection of a higher-dimensional surface.

Despite this, a shared monotonic trend remains visible across all dimensions, consistent with a common geometric origin.

3.7 Multi-Parameter Scaling as Local Chart

The multi-parameter scaling relation [9]

$$\mathcal{C}_{\text{CCI}} \sim \dot{S}_{\text{cg}}^{(+)a}\, I_{\text{nn}}^{-b}$$

can be interpreted as defining local coordinate charts on $\mathcal{M}_{\text{scale}}$. The fitted coefficients show systematic restructuring:

Dimension	a	b	R^2
1D	-1.60	0.33	0.76
2D	-0.08	0.50	0.74
3D	$+0.30$	0.30	0.37

The sign change of a indicates that the local geometry of the manifold changes qualitatively with dimension.

3.8 Physical Interpretation of the Sign Change

The sign change of the entropy coefficient a (from $a < 0$ in 1D to $a > 0$ in 3D) indicates a qualitative shift in the role of entropy production across dimensions.

This sign reversal is consistent with a possible crossover from a noise-dominated regime in low dimensions, where entropy production acts as a destabilising contribution, to a regime in higher dimensions in which dynamical activity may contribute to emergent collective order. This interpretation is speculative and requires further investigation with larger ensembles.

This behaviour may reflect the emergence of collective degrees of freedom in higher dimensions, where activity no longer acts purely as noise but contributes to structured, higher-dimensional dynamics.

Within the scaling-manifold picture, this sign change corresponds to a qualitative change in the local geometry of $\mathcal{M}_{\text{scale}}$: the tangent direction associated with entropy production reverses its orientation relative to the CCI as the system transitions from a low-dimensional to a higher-dimensional regime.

Such a transition is consistent with the emergence of collective degrees of freedom and increased configurational complexity in higher-dimensional systems.

4 Discussion and Implications

4.1 Scope and Limitations of the Manifold Picture

We do not claim that the scaling-manifold picture is unique, but propose it as a minimal geometric interpretation consistent with the observed data. Alternative interpretations may exist, and the present framework should be understood as one possible conceptual organisation of the empirical results rather than a definitive theoretical statement.

4.2 From Scaling Laws to Scaling Geometry

The results suggest a shift in perspective: the entropy–information relation should not be interpreted as a fundamental power law, but as a geometric property of an underlying manifold.

Scaling laws are coordinate-dependent projections of a scaling manifold.

This resolves the apparent breakdown of scaling in higher dimensions: the failure of a single exponent does not indicate a failure of the relation itself, but a limitation of its parametrisation.

4.3 Relation to Structural Free Energy

Within the broader framework [4], the scaling manifold can be interpreted as an observable projection of the structural free energy landscape $\mathcal{F}_{\text{struct}}$:

- $\mathcal{F}_{\text{struct}}$ defines the full high-dimensional selection structure,
- $\mathcal{M}_{\text{scale}}$ represents its projection onto the observable subspace ($\dot{S}_{\text{cg}}^{(+)}$, I_{nn}, $\mathcal{C}_{\text{CCI}}$),
- power-law scaling is a reduced description valid when this projection is effectively one-dimensional.

A possible interpretation in the language of structural free energy is that level sets of $\mathcal{F}_{\text{struct}}$ may project onto the scaling manifold. The exponent degeneracy observed in 3D might then reflect a flattening or saddle-point formation of $\mathcal{F}_{\text{struct}}$ in the corresponding region of configuration space. This remains a tentative picture; establishing this connection rigorously is an open problem.

4.4 Connection to Universality and Critical Phenomena

The scaling-manifold perspective extends traditional scaling concepts:

- universality may extend to geometric structures (manifolds) rather than single exponents,
- effective exponents correspond to local properties of these structures,
- degeneracy reflects the existence of multiple equivalent scaling directions.

4.5 Connection to Known Geometric Structures

The scaling-manifold picture resonates with several established concepts in theoretical physics. In the renormalisation group, the flow of couplings traces trajectories on a manifold in theory space; fixed points and universality classes correspond to geometric attractors on this manifold [11]. Similarly, in Wilsonian effective field theory, integrating out degrees of freedom induces a geometric flow on the space of couplings. The KPZ universality class provides another example in which a fixed-point manifold governs the scaling behaviour of a broad class of nonlinear stochastic systems [12]. The present work proposes an analogous geometric structure for entropy–information scaling in nonlinear field systems, grounded in the same intuition that scaling laws are properties of geometric objects rather than fundamental constants.

4.6 Limitations

The present work is entirely interpretive and empirical. In particular:

- the functional form of F remains unknown,
- the structural parameter ξ is not yet fully identified (though directional MI anisotropy provides a promising proxy),
- the dimensionality of $\mathcal{M}_{\text{scale}}$ is inferred but not rigorously measured,
- the 3D ensemble ($N = 24$) is small; larger ensembles are needed.

A first-principles derivation of the scaling manifold remains an open problem.

4.7 Falsifiability

The scaling-manifold framework leads to several testable expectations:

(i) The width of the effective exponent distribution should increase with spatial dimension.

(ii) The quality of single-exponent fits should systematically decrease with increasing dimension.

(iii) Multi-parameter descriptions should become increasingly relevant in higher dimensions.

For example, one may expect the plateau width $\Delta\alpha$ to increase with spatial dimension, while the quality of single-exponent fits decreases correspondingly. The first direct test of this expectation, presented in Section 3, supports this qualitative picture. The precise functional form of these trends remains an open empirical question at the present stage.

4.8 Outlook

- directional MI diagnostics and extended ξ_{aniso} analysis with larger ensembles ($N \gg 24$),
- extension to 4D and beyond, other field models, and universality class determination,
- numerical estimation of manifold curvature and intrinsic dimension (persistent homology, UMAP, Isomap),
- derivation of F and the multi-parameter family $\mathcal{C}_{\text{CCI}} \sim \dot{S}_{\text{cg}}^{a} I_{\text{nn}}^{-b}$ from the structural free energy,
- connection to Lyapunov spectra, turbulence, and critical phenomena.

5 Conclusion

We have proposed a scaling-manifold framework for understanding entropy–information scaling in nonlinear field systems. The power-law relation $\mathcal{C}_{\text{CCI}} \sim \dot{S}_{\text{cg}}^{(+)}/I_{\text{nn}}^{\alpha}$ is interpreted as a coordinate-dependent projection of an underlying geometric object $\mathcal{M}_{\text{scale}}$.

This framework provides a unified interpretation of:

- robust monotonic scaling across all tested dimensions,
- the dimensional shift of the effective exponent,
- the emergence of exponent degeneracy in three dimensions,
- the sign change of the entropy coefficient in the multi-parameter description.

Preliminary empirical support is provided by PCA of log-space observables, which shows increasing effective dimensionality from 1D to 3D, and by the ξ_{aniso}-split analysis, which demonstrates that directional anisotropy controls the effective scaling structure ($r_s = -0.721$, $p = 0.0001$).

The observed scaling structure reflects not a single universal law, but the dimension-dependent geometry of a scaling manifold.

All results are empirical. A first-principles derivation of the scaling manifold and its geometry remains an open problem.

A Reproducibility

This paper draws on data from the companion papers of this series [5, 6, 7, 8, 9].

All companion code is available at:

https://github.com/Chris4081/structural-selection-principle

Step 1 — Install dependencies:

```
pip install numpy pandas matplotlib scipy scikit-learn
```

Step 2 — Generate the required CSV data files:

```
python cci_alpha_scaling.py        # 1D
python cci_entropy_scaling_2d.py   # 2D
python cci_entropy_scaling_3d_v2.py # 3D with directional MI
```

Step 3 — Generate manifold geometry figures:

```
python manifold_geometry_plot.py
```

Step 4 — Run the ξ_{aniso} analysis:

```
python xi_aniso_full_test.py
```

Output files:

- `paper14_3d_logspace.png` — Figure 1
- `paper14_pca.png` — Figure 2
- `paper14_manifold_evidence.png` — Figure 3
- `paper14_xi_aniso_quartile.png` — Figure 4

Key parameters: Log-transform: $\log_{10}(x + \varepsilon)$, $\varepsilon = 10^{-8}$. PCA: $n_{\text{components}} = 3$. ξ_{aniso}: std/mean of (I_x, I_y, I_z).

B Companion Papers in This Series

Label	Title
[2]	A Structural Selection Principle for Dynamically Consistent Field Configurations
[4]	A Structural Free Energy on the Solution Manifold
[5]	Entropy–Information Scaling of the CCI (1D)
[3]	Landau-Type Theory of Structural Selection
[6]	Entropy–Information Scaling of the CCI (2D)
[7]	Persistence of Scaling in Three Dimensions
[8]	Collapse Analysis Across Spatial Dimensions
[9]	Beyond Single-Exponent Scaling (Multi-Parameter)

All papers available at:

https://kriegchristof.academia.edu/research

References

[1] C. Krieg, *A Structural Selection Principle for Dynamically Consistent Field Configurations*, preprint (2026).

[2] C. Krieg, *A Structural Selection Principle for Dynamically Consistent Field Configurations*, preprint (2026).

[3] C. Krieg, *Landau-Type Theory of Structural Selection in Nonlinear Field Systems*, preprint (2026).

[4] C. Krieg, *A Structural Free Energy on the Solution Manifold*, preprint (2026).

[5] C. Krieg, *Entropy–Information Scaling of the CCI in Nonlinear Field Dynamics*, preprint (2026).

[6] C. Krieg, *Entropy–Information Scaling in Two-Dimensional ϕ^4 Field Dynamics*, preprint (2026).

[7] C. Krieg, *Persistence of Entropy–Information Scaling in Three Dimensions*, preprint (2026).

[8] C. Krieg, *Collapse Analysis of Entropy–Information Scaling Across Spatial Dimensions*, preprint (2026).

[9] C. Krieg, *Beyond Single-Exponent Scaling: Multi-Parameter Entropy–Information Relations*, preprint (2026).

[10] C. Krieg, *Structural Selection — companion code*, https://github.com/Chris4081/structural-selection-principle (2026).

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